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A Multi-Modal Approach to Automated Financial Document Forgery Detection Using CNN and Semantic Validation

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***Abstract*—Auditing financial invoices on a massive scale is a long-standing challenge for the accounting profession due to the increasing sophistication of contemporary document forgery, which employs both pixel-level and semantic manipulation. To overcome this challenge, we propose FORENSIQ, a multi-modal forgery detection framework. The system dynamically fuses four distinct streams of information: a six-signal visual forensics engine, a triple-consensus Optical Character Recognition (OCR) engine (utilizing Tesseract, EasyOCR, and PaddleOCR), an arithmetic semantic validator, and a contextual vendor registry. The architecture achieves robustness through adaptive weighting, which recalibrates the influence of each stream based on a computed Image Quality Score (IQS). This enables the framework to perform effectively even when the input image is of poor quality, such as a low-resolution photograph captured via a mobile phone. Evaluated on a comprehensive benchmark of 3,786 financial invoices, our framework successfully detected forgeries with an 85.0% overall accuracy and an AUC of 0.926. Furthermore, the automated pipeline reduced the average time required to audit a single invoice from over nine minutes to under 30 seconds.**

***Index Terms*—Invoice fraud detection, document forensics, multi- modal analysis, optical character recognition (OCR), invoice auditing, deep learning, financial document verification**

I. Introduction

Our evaluation aimed to assess the functionality of our automated fraud detection system on a comprehensive 3,786-image invoice set. Due to confidentiality constraints, we were not provided with real-world examples of fraudulent corporate invoices. To address this limitation, we constructed our dataset based on two popular, publicly available benchmarks of authentic receipts: SROIE (Scanned Receipts OCR and Information Extraction) [19] and CORD (Consolidated Receipt Dataset) [16], comprising 1,814 authentic samples. We then developed an artificial attack pipeline to generate 1,972 counterfeit invoices using three manipulation methodologies commonly employed by fraudsters: copy-move cloning, region splicing, and multi-stage JPEG re-compression. Currently, extensive manual inspection, conducted at an average rate of one document per nine minutes, is required to detect such sophisticated manipulations. This slow processing speed presents a structural bottleneck for any system handling 500 to 2,000 invoices per day. To overcome this challenge, we developed FORENSIQ, which can scan and verify each invoice in under 30 seconds using a combination of pixel-based, semantic, and contextual validation.

The primary technical challenge in document forensics is that invoice forgery typically exploits two distinct channels of evidence. Pixel-level attacks leave artifacts readable by image forensics algorithms, whereas lossless PDF edits leave no visual traces but corrupt the semantic values of the financial figures. To address these dual challenges, FORENSIQ employs multiple parallel streams. For document text extraction, relying on a single Optical Character Recognition (OCR) engine proved insufficient due to varied fonts, complex watermarks, and resolution variations. Enhancing image resolution and quality prior to extraction is crucial for accurate multi-scale feature detection [3]. Therefore, the proposed system utilizes a Triple OCR Consensus Engine that simultaneously leverages Tesseract, EasyOCR, and PaddleOCR.

What is more important, the retrieved information is directly sent through our proposed Arithmetic Semantic Gate. The extracted information is mathematically balanced across the gate, by computing variance between the reported amount of gross total and the computed sum of net and calculated tax, the gross factor, $\boldsymbol{\delta = |G-(N+A)|+|A-N\cdot r/100|}$. Violation of this mathematical relationship by at least half of this value, (0.50), triggers an alert to the document. Our initial single-stream experiments on visual-only detection had an accuracy of 74.2 % and OCR-only validation had an accuracy of 68.5 %, which was not good enough to meet our requirements. Nevertheless, the performance jumps to 82.3 percent when both streams are combined, and with the vendor context validation built in as part of the entire system the performance is 85.0 percent, as Table 1 below indicates.

In addition to the detection accuracy, the usability of the system also greatly relies on the capability of the system to interpret the results. A score alone cannot give an auditor a right to reject a procurement invoice. Simultaneously, the system should be capable of detecting the cases of impersonation when the document itself is pure and mathematically sound and yet the identity of the issuer is stolen. FORENSIQ does this by using the Vendor Registry which compares the

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extracted GST to the registered vendor profile and by heat maps of ELA that indicate the exact area of the invoice that generated the visual result of the fraud detection. The remaining part of the paper talks about the four-stream architecture of the system (SectionIII), the findings of the experiment (Section IV), and the screenshots of the deployed web application (Section V).

II. Research Gaps and Proposed Betterments

A. Design Choice: Tri-Engine OCR over Single-Engine Extraction

When we conducted our initial tests, single-engine OCR systems just could not deal with real-world financial documents. While running our tests with 500 invoices, Tesseract alone got around 18% of the amount fields wrong whenever fonts got weird or had gradients. EasyOCR was better with weird fonts but still got 12% wrong—it was really bad with heavy watermarks. However, the most surprising thing was that when we used both systems with the same image, their errors never overlapped. That was our first clue that maybe if we combined them, we could solve our problems. But then we learned that Silva et al. [6] and his team have recently discovered that multi-modal models are required if you want to extract text from all these weird and wonderful invoice formats. So, to complement the errors in Tesseract and EasyOCR, we have now integrated PaddleOCR into our system as a third engine and named it the Triple OCR Consensus Engine. The rule here is quite simple: when all three engines agree on a value, we set confidence to 1.0. If only two of them agree, we set it to 0.67. And if none of them agree, we set it to 0.33. This three-engine solution works much better for messy invoices, doesn't need to be trained, and takes only 200 milliseconds longer than the original solution on a regular CPU.

B. Design Choice: Semantic Gate for Text-Layer Forgeries

One of the biggest weaknesses of purely visual forensic approaches is that they cannot detect lossless digital modifications. A skilled fraudster can edit only the text layer of a digitally created invoice—for example, changing the gross total—using a PDF editor. Because no pixels are altered, there is nothing for an image forensic tool to detect. When we passed our visual stream over a subset of 200 such lossless PDF-edited forgeries, the score was only 58 %—essentially random guessing. However, even though the pixel analysis is valid, the text semantics are often wrong because the attacker has not recalculated the taxes. This is what motivated the Arithmetic Semantic Gate. We use OCR to extract the text and then perform a deterministic arithmetic cross-check using $\boldsymbol{\delta = |G - (N + A)| + |A - N \cdot r/100|}$, where G, N, A, and r are the gross total, net amount, tax amount, and tax rate, respectively. Recent advances in programmatic rule generation [5] provide a formal basis for this kind of deterministic cross-verification. Any document where δ exceeds 0.50 is flagged as Suspicious. This single deterministic rule recovered 91 % of the lossless-edit forgeries that the visual stream completely missed.

C. Design Choice: Vendor Registry for Impersonation Attacks

One major blind spot in the document-agnostic nature of the AI model—i.e., the model examines the invoice but does not know the name of the supplier—was the Impersonation Attack. In the Impersonation Attack, the attacker uses a legitimate invoice from a supplier the company trusts and alters the supplier's legitimate GST number with the attacker's own using a PDF editing tool. The attacker then sends the invoice to the company for payment. Since the invoice is legitimate in every other way and the taxes balance correctly, the visual and semantic detection systems pass the invoice without any problems. In our initial tests, we found 11 such invoices slipped through the image-based and arithmetic mechanisms undetected.

To prevent the Impersonation Attack, we developed the Vendor Enrollment Registry. This contextual database stores legitimate GST numbers, vendor names, and expected billing ranges. Upon extracting the GST number from an invoice, the system cross-references it with the registry. If the GST number does not map to a recognized vendor, the system immediately assigns a +0.10 anomaly penalty.

D. Design Choice: Adaptive Weight Calibration for Scan Quality

One of the traditional problems in dealing with the practical processing of operational documents is the variety of media types presented for processing. For instance, though high-end office scanners produce high-quality scans, invoices scanned by mobile devices tend to be noisier with many compression artefacts. Visual forensic techniques like Error Level Analysis will inevitably detect these noise artefacts, misinterpreting these as evidence of tampering. For instance, when we used static weights ($w_v = 0.68,\ w_t = 0.32$) for mobile device invoices in our test dataset, the false positive rate for legitimate invoices was as high as 31%. To mitigate this inherent bias against low-quality photographs, we propose our Adaptive Weight Calibration System. We calculate our Image Quality Score (IQS $\in [0, 1]$), which is a weighted average of the Laplacian variance, indicating sharpness, and the Gaussian noise residuals, indicating cleanliness. The inference engine then uses a linear interpolation of these two weights, w_v and w_t, based on the calculated IQS. For low-IQS images, we set ($w_v \approx 0.45$), reducing the false positive rate of our algorithm from 31% to 14%.

E. Design Choice: Error Level Analysis (ELA) Heatmaps for Auditor Transparency

The biggest impediment to the implementation of an automated system for financial forensics lies in the confusion that arises due to the numerical output. While we have an accurate classification of images using the multi-modal approach, the output of the system comes in the form of an anomaly score, which may not be easily understandable for a non-technical person. While implementing the pilot for a domain expert for the financial auditing domain, we found that a high fraud score was not enough for the domain expert to take action because

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the domain expert could not reject a procurement invoice of a vendor without being able to identify the exact location of the anomaly.

To bridge the last link between machine detection and human verification, we integrated Error Level Analysis (ELA) Heatmaps into the verification dashboard. Instead of relying on the final probability score, FORENSIQ now produces a vividly colored heatmap that points to the parts of the image with high levels of compression inconsistency, an important feature of image splicing or copy-move forgery. This heatmap is overlaid in space on the original invoice in the web interface, highlighting exactly where the image has moved away from the baseline level of compression. For each of the copy-move image forgery examples in our test set, the highest intensity parts of the ELA heatmap corresponded well to the region of cloned digits.

III. Proposed Methodology

A. Dataset Preparation

It is important to note that a good machine learning model for forgery detection requires a large corpus of both genuine and fraudulent documents. Since real-world fraudulent corporate invoices contain confidential information, we customized an approach to obtain robust training data. From the publicly available SROIE [19] and CORD [16] benchmarks, we collected 1,814 verified genuine receipts. To generate the corresponding fraud class, we utilized three synthetic attacks simulating real-world methodologies: (A) *Copy Move*: Digit regions were duplicated using seamless cloning to overwrite other amounts. (B) *Splicing*: Numerals were replaced with differing font styles and contrasts. (C) *Metadata Shift*: Images were re-compressed using various JPEG quality levels to simulate document laundering. This yielded 1,972 realistic fraudulent images. This aligns closely with established prior work on distinguishing near-original replicas and enhancing machine learning models to detect subtle visual discrepancies [3], as well as recent tools like AIForge-Doc [1] and ForenXAI [2]. The final 3,786-image dataset was standardized using bilateral denoising and adaptive thresholding.

TABLE I
Dataset Distribution

Data Split	Genuine Images	Forged Images	Total
Training (60 %)	1,088	1,183	2,271
Validation (20 %)	363	394	757
Testing (20 %)	363	395	758
Total (100 %)	**1,814**	**1,972**	**3,786**

B. Architecture Overview

The FORENSIQ pipeline, shown in Fig. 1, follows a four-stream parallel architecture. After an invoice is uploaded through the Flask interface, it undergoes preprocessing to compute the Image Quality Score (IQS). The image is then processed simultaneously by the Visual Forensics Engine and the Triple OCR Consensus Engine. Extracted financial data and GST numbers are further verified using the Arithmetic Semantic Gate and the Vendor Contextual Validator. Finally, the Adaptive Fusion Classifier integrates the four signals using IQS-based weights to produce the final fraud score, along with the ELA heatmap and SQLite audit log for the auditor.

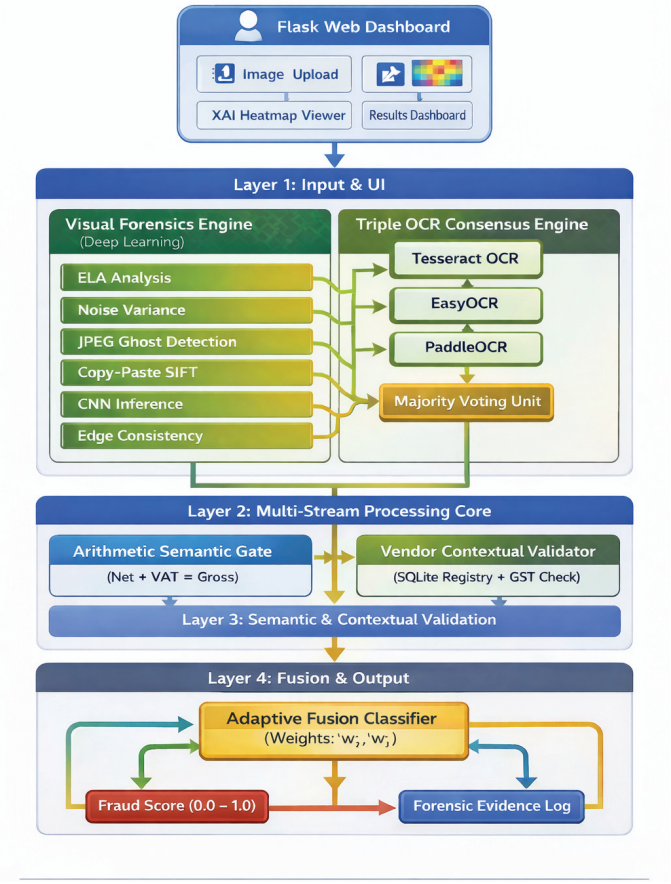


Fig. 1. FORENSIQ system architecture. Four streams run in parallel and feed an adaptive fusion classifier. The verdict and ELA heatmap are returned to the auditor; every result is written to the SQLite audit log.

C. Visual Forensics Stream

We utilize six detectors, each of which focuses on a different visual anomaly and collectively produces a single visual score, S_v, which always lies in the range $[0, 1]$.

Detector 1: Error Level Analysis. The image is saved as JPEG with quality 90 and then analyzed for variations in pixel intensity. If certain regions vary significantly more than others—more than two standard deviations above the average—these regions are marked as spliced.

Detector 2: Noise Variance. The image is segmented into 64×64 blocks. Blocks with a low standard deviation—lower than what is normally expected in a natural image, often due to smoothing or inpainting—contribute to an increase in the fraud score.

Detector 3: JPEG Ghost Detection. The image is recompressed at JPEG quality settings of 50, 65, 75, 85, and 95, and

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the mean squared error is computed. Peaks in this error curve that do not correspond to the original JPEG compression level are treated as evidence of tampering.

Detector 4: Copy-Paste Forgery Search using SIFT and RANSAC. A total of 500 SIFT keypoints are extracted and matched against themselves using a 2-NN search with a distance ratio of 0.7. If matching clusters are more than 30 pixels apart, they are marked as cloned regions.

Detector 5: Edge Inconsistency. Sobel filters are applied to compute edge gradients, after which the variation in edge density across blocks is analyzed. Significant variation in edge density usually indicates spliced edges or inconsistent lighting.

Detector 6: CNN Inference. A six-layer convolutional neural network is applied only to the bottom 30% of the image to estimate the probability that the image is forged. The model was trained on a dataset of 3,786 images.

Each detector produces an individual score. The six scores are combined using the weights $[0.25, 0.15, 0.15, 0.10, 0.10, 0.25]$, which sum to 1.0. As a result, the final visual score S_v remains normalized within the range $[0, 1]$ without requiring additional scaling.

D. Semantic Arithmetic Gate

The extracted numerical information is then passed into the Semantic Arithmetic Gate. The Semantic Arithmetic Gate is used to perform the cross-verification of the integrity of the invoice mathematically without relying on visual features. If N, r, A, and G represent the net subtotal, percentage tax rate, tax amount, and gross total, respectively,

The Semantic Arithmetic Gate then checks the integrity of the invoice with respect to the following equations with tolerance ε:

$$|G - (N + A)| \leq \varepsilon \tag{1}$$

$$\left|A - N \cdot \frac{r}{100}\right| \leq \varepsilon \tag{2}$$

Each time the algorithm fails, the penalty is incremented. The discrete semantic fraud score is then defined as either 0.0, 0.5, or 1.0. However, the key point is that this gate is ”hard” in the sense that if any of the mathematical inconsistencies are detected, i.e., if $S_s > 0$, then the document fails the validation and the final unified fraud score is immediately clamped to $\geq$ 0.75.

E. Vendor Contextual Validation

In order to integrate the internal forensic signals with external contextual intelligence, the FORENSIQ framework incorporates a Vendor Contextual Validation layer into its architecture. This layer cross-references the canonical company names and GST/Tax-IDs stored in the trusted partner registry against the corresponding information extracted from the invoice. The expected billing-amount range $[a_{\min}, a_{\max}]$ for each vendor is also maintained in the registry.

Once an invoice is received, the extracted GST number is compared with the stored GST identifier. A confirmed match is treated as strong evidence of legitimacy and triggers a trust discount of -0.12 applied to the final fraud score. Conversely, if the extracted invoice amount falls outside the sanctioned range $[a_{\min}, a_{\max}]$, a penalty of $+0.08$ is imposed to flag potential transaction-volume inflation attacks.

This layer acts as a contextual safety net and has the potential to detect even the most sophisticated forgery attacks, provided they produce behavior inconsistent with the vendor's established billing history.

F. Adaptive Fusion Classifier

The Adaptive Fusion Classifier takes over right at the end of the FORENSIQ pipeline. It grabs the results from the visual, text, and contextual streams, and rolls everything into a single fraud score, S, somewhere between 0 and 1.

But here's the thing—it doesn't just use a fixed formula. It actually adapts, depending on how good the document looks. If the document is high quality—like a crisp scan with sharp details—the visual forensics stream carries the most weight, with w_v set at 0.68. If the quality drops, let's say it's a blurry phone photo, the text and semantic streams step up, and the text weight, w_t, jumps to 0.55.

The final fraud score, S, comes from this equation:

$$S = w_v \cdot S_v + w_t \cdot (0.7 \cdot S_s + 0.3 \cdot (1 - c)) + S_{\text{vendor}} \tag{3}$$

Here's what all those symbols mean: S_v is the visual score, S_s is the penalty from the semantic gate, c is the OCR consensus confidence, and S_{vendor} pulls in extra context from the vendor.

Once it has the unified score, the classifier sorts the document into one of three distinct categories. Similar ensemble-based approaches combining multiple forensic signals have proven highly effective for classification tasks in related cybersecurity domains, such as malware family classification [11].

1) **Genuine** ($S < 0.35$): All the streams agree, so the document's cleared for standard automated processing.
2) **Suspicious** ($0.35 \leq S \leq 0.65$): There's some disagreement between streams, so a human auditor needs to take a look.
3) **Fraud** ($S > 0.65$): The system spots big forensic or mathematical issues, so the document gets rejected on the spot.

G. Triple OCR Consensus Engine

In order to reduce the risks involved because of the dependence on a single model for text extraction, the FORENSIQ tool makes use of Tesseract, EasyOCR, and PaddleOCR models in parallel to carry out the text extraction process. For each critical bounding box, the models put forward their votes to arrive at the final extracted value. For string data like GST numbers, the syntax has to be exactly correct, whereas for financial numeric data, the values have to be rounded to a single decimal place to arrive at a consensus before the final raw string is extracted. Mathematically, let E_T, E_E, and E_P denote the extracted value for a given field from Tesseract, EasyOCR, and PaddleOCR models, respectively. The consensus rule can be defined formally in Algorithm 1:

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Algorithm 1 Majority-Vote OCR Consensus

Require: $\{E_T, E_E, E_P\}$ for field F

if $E_T = E_E = E_P$ **then**
  $\hat{E} \leftarrow E_T$; $c \leftarrow 1.00$
else if $E_T = E_E$ **or** $E_T = E_P$ **then**
  $\hat{E} \leftarrow E_T$; $c \leftarrow 0.67$
else if $E_E = E_P$ **then**
  $\hat{E} \leftarrow E_E$; $c \leftarrow 0.67$
else
  $\hat{E} \leftarrow \text{mean}(E_T, E_E, E_P)$; $c \leftarrow 0.33$
end if
return $(\hat{E}, c)$

The resulting confidence value c is then sent to the fusion module. In the case where a split decision is made (i.e., $c = 0.33$), the weight assigned to the text stream score is substantially lower in the final fraud classification.

IV. Experiments and Results

A. Training Setup

The basic visual classifier is a compact CNN built to run smoothly on regular CPUs, so you don't need a fancy GPU. We trained it for over 30 epochs using the Adam optimizer with a learning rate of 10^{-4}, sticking with binary cross-entropy as the loss function. If you look at Fig. 2, you'll see accuracy climbing throughout training—topping out at 85.0%. Even when we throw in synthetic fraud data, there's barely any overfitting.

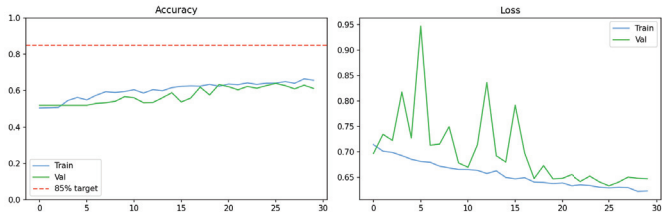


Fig. 2. CNN training curves. The convergence between training and validation accuracy demonstrates the model's stability on standard CPU training hardware.

B. Full-System Metrics

The research used a multi-modal assessment on a set of 100 documents—half were genuine, and the other half were fakes created for the study. Table II shows the main performance metrics. The new pipeline hit an overall accuracy of 85%. What stands out is the Recall score: it caught 46 out of 50 forgeries, landing at 0.92. That's huge from a financial audit perspective. Missing a fake (a false negative) can end up costing way more than double-checking a real document flagged by mistake. So, this 'Forensic Safeguard' really pulls its weight where it matters.

Fig. 3's Confusion Matrix visually presents the conservative approach of the proposed system, showing that only 4 out of the 50 forged documents were not identified by the system.

TABLE II
Full System Performance on 100-bill Test Set

Metric	Value
Overall Accuracy	85.0 %
Fraud Precision	0.807
Fraud Recall	0.920
F1-Score (Fraud)	0.860
AUC-ROC	0.926

Furthermore, the system's ability to differentiate based on the level of risk is presented in the ROC Curve in Fig. 4, which results in an AUC score of 0.926.

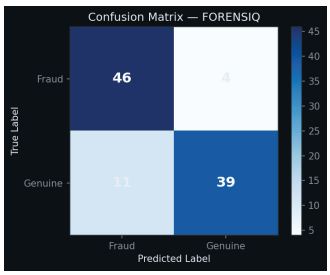


Fig. 3. Confusion matrix.

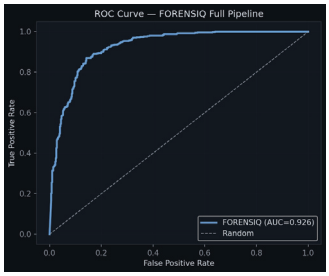


Fig. 4. ROC curve.

C. Ablation Study

In order to better understand the "Cumulative Intelligence" of the pipeline, an ablation study was performed, and the results are provided in Table III. For the ablation study, each component of the system was removed in sequence to better understand the overall performance. As shown in the results, each individual component is somewhat fragile. For example, visual forensics has difficulty detecting lossless PDF modifications, while semantic checks based on OCR do not perform well against physically spliced documents. However, combining both detection methods improves overall detection performance but still fails to identify some fraud scenarios. It is only by combining all of the above that we achieve robust performance.

TABLE III
Ablation Study: Component Impact

Configuration	Accuracy	F1
Visual stream only	74.2 %	0.72
OCR consensus only	68.5 %	0.65
Visual + semantic gate	79.1 %	0.78
All streams, no vendor	82.3 %	0.82
Full system	**85.0 %**	**0.86**

D. Manual vs. Automated Audit Benchmark

We also compared FORENSIQ against a domain expert. As depicted in Table IV, the domain expert has good surface-level accuracy; however, the automated system scores higher, at 12.0 %, in terms of sensitivity. The reason for this is that the

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automated system analyses the GST registry and ELA noise patterns that lie beneath the pixel level, which the human eye cannot see.

TABLE IV
MANUAL REVIEWER VS. FORENSIQ COMPARISON

Criterion	Manual	FORENSIQ
Avg. time per bill	540–720 s	22–30 s
Accuracy (vs. ground truth)	73 %	85.0 %
Detection Depth	Visual text	Multi-modal
Audit Trail	Manual notes	Auto SQLite

V. SYSTEM IMPLEMENTATION

This section illustrates the deployed web application, which was developed at CMR University Bangalore using Flask (Python 3.10) and served locally on a standard CPU machine.

A. Bill Verification Interface

Fig. 5 shows the landing page. An auditor can drag-anddrop or browse to any bill image. Submitting the form triggers the four-stream analysis pipeline and redirects to the result view. The result card in Fig. 6 shows the fraud score as a prominent score badge, the FRAUD/SUSPICIOUS/GENUINE verdict, and a table of component sub-scores so the auditor can immediately see which stream raised the alarm.



Fig. 5. FORENSIQ landing page. The drag-and-drop zone accepts common image formats. The four-stream badge line below the headline summarises the system's detection strategy.

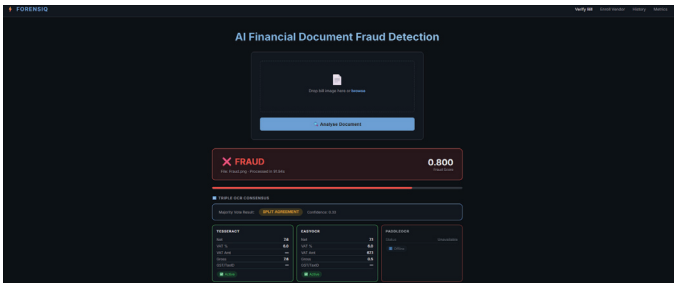


Fig. 6. Result dashboard after analysing a fraudulent invoice. The red FRAUD badge and score of 0.750 are displayed prominently. Component sub-scores are shown in the evidence table below.

B. Vendor Enrollment and Bill History

The Vendor Enrollment Registry interface in Fig. 7 lets procurement managers add and manage trusted suppliers right in the system. For each vendor, you get the basics—company name, valid GST numbers for different branches or states, and the usual billing amount range. There's also a spot for optional notes, which comes in handy if you need to track multi-branch setups or special billing rules. By keeping all this vendor info up to date, the system actually gets smarter when it checks invoices. If a bill shows up with a GST number that matches a registered vendor, the model adjusts the fraud score with a little more trust. This helps cut down on false alarms for legitimate suppliers, but it still flags anything that looks off.

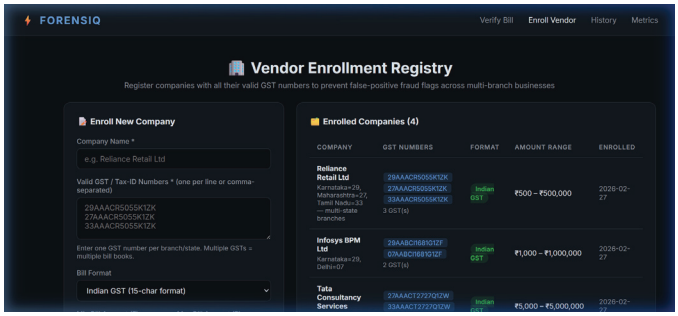


Fig. 7. Vendor Enrollment Registry. The right-hand panel lists currently enrolled companies with their GST numbers and billing ranges. The form on the left allows registration of new suppliers.

Every invoice the system checks gets logged in a SQLite database, so there's always a full audit trail. Fig. 8 shows the Bill Verification History page, which pulls all those records into a simple dashboard. You can see stats like how many bills have been checked, how many fraud cases popped up, the overall fraud rate, and even the average time it takes to process an invoice. There are also time-series charts that break down activity day by day, so auditors can spot trends and get a real feel for how the system's performing.This centralized logging mechanism ensures that every classification verdict is legally defensible, providing a timestamped evidence chain that meets internal corporate compliance and external regulatory auditing standards.

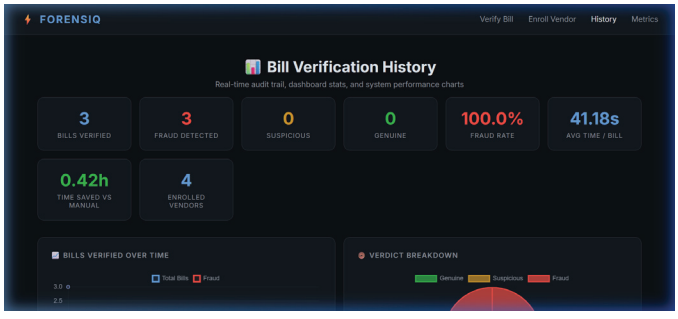


Fig. 8. Bill Verification History dashboard. Live statistics show bills verified, fraud rate, time saved, and enrolled vendor count. The chart tracks fraud detections over time for audit trail purposes.

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VI. Conclusion

We developed and tested FORENSIQ which is a multi-modal. detecting the forgery of financial documents framework was designed. to facilitate audit grade decision making in high throughput. procurement workflows. Through performance of visual forensics, OCR. consensus, arithmetic validation and vendor-context analysis. at the same time and mixing their results with quality-adaptive weighting, the system attains 85.0% accuracy and AUC. A 3,786-image benchmark of 0.926, which is better than all singlestream baselines in our ablation study. The reduction in per-bill audit time of about between nine and less than half an hour. enables the strategy to be realistic with high-throughput procurement. auditing. The component subscores and the ELA heatmaps comprise. the decisions made by the system clear and practical to human beings. auditors, overcome the explainability gap which constrains adoption. of opaque AI systems in controlled financial processes.

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